

Position as a Variable, Not a Label: Tactical Distance and Role-Conditioned Performance in Versatile Footballers

Soccer

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Abstract.

Versatile footballers are routinely shifted across full-back, midfield, and attacking roles, yet flexibility does not mean every role suits the same player equally well. Most football datasets treat position as a fixed label, obscuring what changes when the same player is redeployed. We model position as a time-varying tactical role signal and ask whether tactically larger role changes are associated with larger within-player performance-profile shifts. The aim is to describe where versatility produces a stable role advantage, a trade-off, or an effectively neutral profile.

Position is a variable, not just a label

Tactical distance links role changes to changes in the same player's performance profile

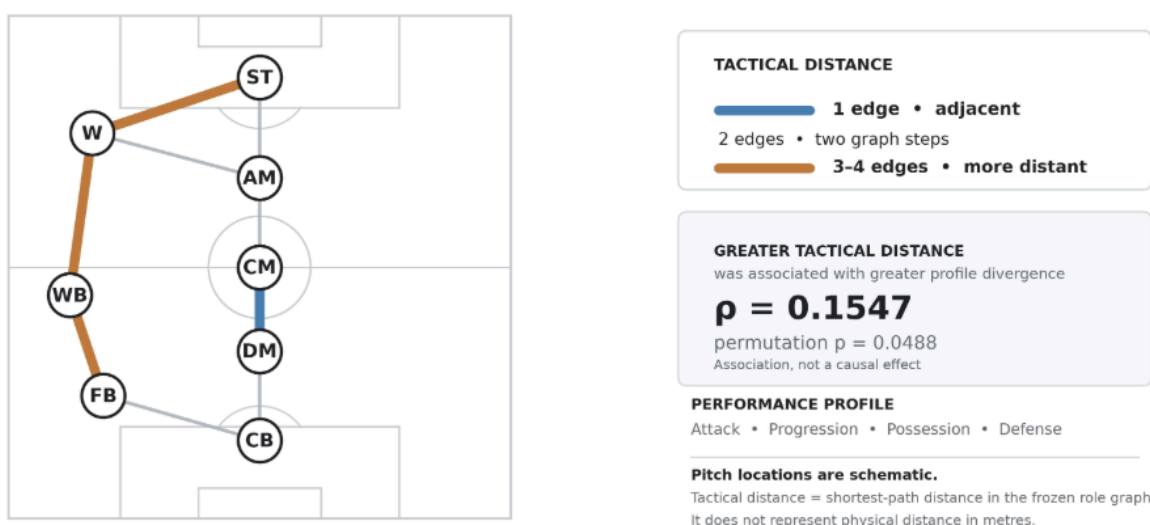


Figure 1. Tactical role graph for eight outfield position families. Pitch locations are schematic; tactical distance is shortest-path distance in the frozen conservative graph, not physical distance in metres. Greater graph distance was associated with larger within-player performance-profile divergence ($p=0.1547$; permutation $p=0.0488$)

Using SoccerSolver formation-segment timing linked to player-match statistics, we constructed a Player $\times$ Match $\times$ dominant-position-family dataset from 23,627 player-match observations, covering 120 versatile outfield players and 159 within-player role comparisons after requiring $\geq 70\%$ role purity, ≥ 600 clean minutes, and ≥ 8 qualifying matches per role. Each match was assigned one dominant family among CB, FB, WB, DM, CM, AM, W, and ST. Outputs were normalized per 90, standardized within role-family reference groups, and summarized as Attack, Progression, Possession, and Defense scores. Positional change was measured as Euclidean divergence across those four score differences. Sixteen comparisons (10.06%) showed stable advantage, 36 (22.64%) a trade-off, 40 (25.16%) neutrality, and 67 (42.14%) insufficient evidence. Baseline cross-family transitions exceeded adjacent transitions by 0.1873 (95% CI [0.0583, 0.3285], $d=0.8939$). The conservative estimate remained directionally positive. Distance was positively associated with divergence ($p=0.1547$; permutation $p=0.0488$). Excluding disputed classifications produced +0.1854 (95% CI [0.0565, 0.3241]); no leave-one-player or leave-one-transition check reversed the direction.

Three Archetypes of Versatility

Frozen Phase 4.5 case studies show stable advantage, trade-off, and near-neutral role profiles

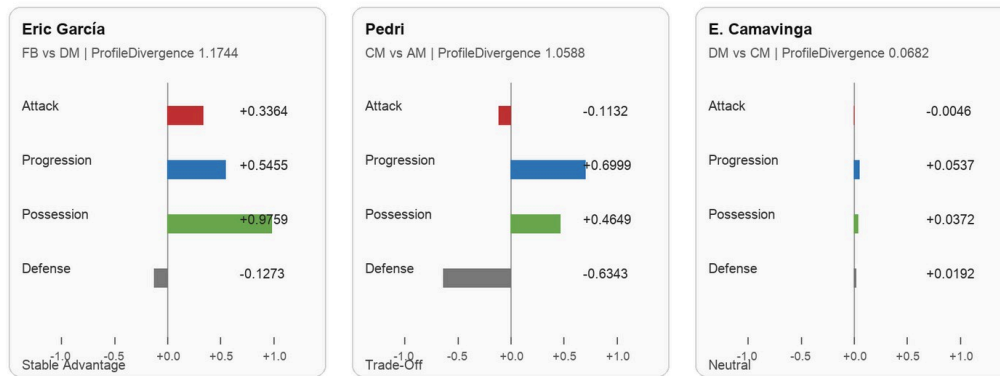


Figure 2. Stable advantage, trade-off, and near-neutral versatility remain visible in the frozen archetype examples.

Eric García illustrates stable FB–DM separation, Pedri a CM–AM trade-off, and E. Camavinga a near-neutral DM–CM profile. The framework can inform selection, recruitment, and squad planning without automating decisions. Applied within one club, it can compare how its versatile players performed across roles they actually played. Because the data contain formation-segment timing and player-match totals but no event timestamps, coordinates, or tracking, results are match-level associations rather than causal effects or event-level positional attribution.

This research was developed in collaboration with SoccerSolver. SoccerSolver currently works with more than 10 football clubs.

Open-source repository: <https://github.com/RupayanHalder39/MITSloanProblem12-Football-Position-Versatility>